ECO 646 Macroeconomics: Unit 4 Project

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For this project, you will analyze US monetary policy through the lens of various monetary policy rules. Relevant data can be accessed from the FRED database. Use data at a quarterly frequency for Questions 1 and 2, and data at an annual frequency for Question 3. Note that the most recent quarter for which data is fully available is 2025Q2.[1] Use averages as the aggregation method for both frequencies.

1. Current Monetary Policy

a) Consider the simple monetary policy rule presented in the text: $R_t = \bar{r} + \bar{m}(\pi_t - \bar{\pi})$. We want to use this equation to estimate the federal funds rate: $FFR_t = \bar{r} + \bar{m}(\pi_t - \bar{\pi})$. We can do so in the following way:
   - First, we need an appropriate neutral interest rate $\bar{r}$. The current best estimate is $\bar{r} = 0.5\%$.
   - Second, we need a measure for current inflation π_t for 2025Q2. The Fed's preferred inflation index is the Personal Consumption Expenditures price index (series PCEPI on FRED). Measure the 2025Q2 inflation rate as the PCE price index percentage change from a year ago.
   - Third, we need an inflation target $\bar{\pi}$. The Fed's stated inflation target is 2%.
   - Fourth, we need a value for the inflation sensitivity parameter $\bar{m}$. A commonly imposed value is $\bar{m} = 1.5$, which indicates that the Fed will respond more than one-for-one to a given increase in inflation.

[1] Normally, Q3 data would be available by now. However, due to the government shutdown, releases have been delayed, so the most recent data we have are for 2025Q2.

With the above assumptions, the simple monetary policy rule can be operationalized as:

$$FFR_t = 0.5 + 1.5 \times (PCE\ inflation_t - 2\%).$$

What does the simple rule indicate the fed funds rate should be in 2025Q2?

b) Taylor (1993) was the first to propose a modern monetary policy rule (MP rules are also commonly referred to as the Taylor rule). Taylor's original rule accounted for short-run output $\tilde{Y}_t$ in addition to inflation:

$$FFR_t = \bar{r} + \bar{m}(\pi_t - \bar{\pi}) + \bar{n}\,\tilde{Y}_t.$$

- Short-run output can be calculated as a FRED series and accessed here.
- In Taylor's original formulation, he selected an output sensitivity parameter $\bar{n} = 0.5$.

What does Taylor's original rule indicate the fed funds rate should be in 2025Q2?

c) Commentators such as former Fed Chair Ben Bernanke have suggested that the Fed has responded more strongly to short-run output in recent years, implying $\bar{n} = 1$ might be a more accurate value for the output sensitivity parameter.

What does Taylor's rule with $\bar{n} = 1$ indicate the fed funds rate should be in 2025Q2?

d) One interpretation of the average inflation targeting regime implemented in August 2020 is that the Fed is putting relatively less weight on inflation when making policy decisions. In that case, it may be more accurate to lower the value of the inflation sensitivity parameter to $\bar{m} = 1$.

What does Taylor's rule with $\bar{n} = 1$ and $\bar{m} = 1$ indicate the fed funds rate should be in 2025Q2?

e) Which of the four rules from parts (a)–(d) do you think is closest to the Fed's current operating procedure? Explain your answer.

2. Past Monetary Policy

a) Download the PCE inflation rate (as percentage change from a year ago), short-run output, and the federal funds rate (series DFF) from FRED for 1994Q1 through 2019Q4. Simulate a counterfactual fed funds rate according to each of the four monetary policy rules from Question 1 (simple rule, original Taylor rule, Taylor rule with $\bar{n} = 1$, Taylor rule with $\bar{n} = 1$ and $\bar{m} = 1$) over this time period. Set $\bar{r} = 2\%$ rather than equal to 0.5% since that is a more accurate description of the neutral interest rate over this period. Make a plot containing the actual federal funds rate and the four counterfactual rates based on the policy rules.

Which rule appears to describe the historical evolution of the federal funds rate most accurately? (Hint: if it is not clear from eyeballing the plot, consider calculating correlations with the actual fed funds rate.)

b) Next, estimate historical values of the sensitivity parameters $\bar{m}$ and $\bar{n}$. You can do so by regressing the fed funds rate minus the neutral interest rate ($FFR_t - 2\%$) on π_t and $\tilde{Y}_t$ without a constant/intercept. The coefficient on π_t is an estimate of m̄ and the coefficient on $\tilde{Y}_t$ is an estimate of $\bar{n}$.

Report your estimates of the two parameters. Are these consistent with the monetary policy rule that appeared most accurate in part (a)?

c) Compare your responses to parts (a) and (b) of this question with your response to part (e) of Question 1.

3. Future Monetary Policy

a) The Fed releases a summary of forecasts for real GDP growth, the unemployment rate, inflation, and the fed funds rate four times a year in a document called the Summary of Economic Projections (SEP). The latest SEP was released in September and can be accessed here. For this question, focus on the first four columns of Table 1, which provide median projections for 2025, 2026 and 2027. We want to see if the Fed's internal projections are consistent with the monetary policy rule that you selected as most accurately describing current policy in part (e) of Question 1.

Write down the equation form of that monetary policy rule here (with $\bar{r} = 0.5\%$).

b) The SEP gives direct forecasts of PCE inflation. These can be inserted as π_t in your monetary policy rule (note that projections in the previous June 2025 SEP are listed below the current September projections — you can ignore these). The SEP does not directly forecast short-run output however, so you will have to back it out from the real GDP growth forecast.

- The 2025 forecast states that the Fed projects real GDP to grow 1.6% from 2024 to 2025. The 2024 value of real GDP can be found via FRED (series GDPC1). Multiplying the 2023 value by (1 + 1.6%) will give an estimate of 2025 real GDP. Similarly, the 2025 SEP forecast projects real GDP to grow by 1.8% from 2025 to 2026, so the level of 2026 real GDP can be calculated by multiplying the projected 2025 value by (1 + 1.8%). Real GDP can be calculated for all three years in this manner.
- FRED offers estimates of potential real GDP $\bar{Y}_t$ through 2035 (Series GDPPOT). Note that these estimates come from the Congressional Budget Office (CBO) rather than the Fed.
- Short-run output for 2025–2027 can then be calculated as:

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t},$$

and inserted into your monetary policy rule.

What does your monetary policy rule indicate the fed funds rate should be in 2025, 2026 and 2027?

- The Fed's internal projections for the fed funds rate are presented in the second-to-last row of Table 1 (for example, a median projection of 3.6% for 2025).

  Are the Fed's projections consistent with the monetary policy rule?

- **How do you think the Fed's internal projections of potential real GDP ($\bar{Y}_t$) might differ from the CBO's?** Explain how this could influence the estimate from your monetary policy rule.